2. T. Chan, *Unix System Programming Using C++*, 1st ed. Prentice-Hall, Inc., 1996

3. J. B. Maurice, *The Design of the UNIX Operating System*, 1st ed. Pearson, 1986

SWE 4537 Server Programming Credit: 3.00

Web service, HTTP protocols, IP, port, URL, routing. Web security fundamentals, CORS, authorization, authentication, OAuth, Social authentication, SSO. Database Connectivity, ORM. State Management, Session, Cookie, WebSocket, Server Push. Performance, scaling, load balancing, lazy loading, caching;

Fundamentals of web service deployment, Background service, security and firewall, multi-threading.

SWE 4533 Cryptography Credit: 3.00

Fundamentals: OSI security architecture, Security goals, Types of attacks, Cryptography and Cryptanalysis basics, Steganography, Classical encryption techniques, Cipher principles ;

Private/Shared/Symmetric Key Cryptography: Data Encryption Standard (DES), Block cipher design principles and modes of operation, Evaluation criteria for AES, AES cipher, Triple DES, Placement of encryption function, Traffic confidentiality, Key management, Key Distribution Center (KDC);

Public/Asymmetric Key Cryptography: Key management, Diffie Hellman key exchange, Elliptic curve architecture and cryptography, Introduction to number theory, Confidentiality using symmetric encryption, Public key cryptography and RSA, Theory: Euclidean algorithm, Euler Theorem, Fermat Theorem, Totient functions, multiplicative and additive inverse, Public Key Infrastructure (PKI), PKI Trust Models, Certificate standard (PKIX and X.509, Certificate authority (CA), Certificate revocation.

Hash Function: Hash functions, Security of hash functions and MACS, MD5 Message Digest algorithm, Secure hash algorithm (SHA), HMAC digital signatures, Digital signature standard, Elliptic Curve Cryptography (ECC)

Recommended Texts:

1. D. R. Stinson and M. Paterson, *Cryptography: Theory and Practice*, 4th ed. CRC Press, 2018

2. W. Stallings, *Network and Internetwork Security: Principles and Practice*, 6th ed. Pearson, 2013

3. B. A. Forouzan, *Cryptography & Network Security*, 2nd ed. McGraw-Hill, Inc., 2011

SWE 4535 Game Development Credit: 3.00

Introduction to the Class, Role of the Game Designer, Formal elements of games, Dramatic elements of games and Narrative Design, System dynamics, Challenge, Skill and Chance, Conceptualization, Communication, Social Play, Games as Culture, Game Economics, Level design and properties of living things, Functionality, Completeness and Balance, Simple Playtesting and Quality Assurance, Game Project.

Recommended Texts:

1. J. G. Bond, *Introduction to Game Design, Prototyping, and Development: From Concept to Playable Game with Unity and C#*, 3rd ed. Addison-Wesley Professional, 2022

2. T. Fullerton, *Game Design Workshop: A Playcentric Approach to Creating Innovative Games*, 5th ed. A K Peters/CRC Press, 2024

3. B. Brathwaite and I. Schreiber, *Challenges for game designers*, 1st ed. Charles River Media, 2008

SWE 4532 Network Programming Lab Credit: 0.75

Sessional works based on SWE 4531.

SWE 4538 Server Programming Lab Credit: 0.75

Sessional works based on SWE 4537.

The students will incrementally deliver a web service that can be consumed by many different clients over the network. The requirements of the service will be given week-by-week, the delivery will also be in a weekly interval. The service will be tested with API client applications like postman.

Suggested Technologies: .NET/Java/Node/Python.

SWE 4534 Cryptography Lab Credit: 0.75

Sessional works based on SWE 4533.

SWE 4536 Game Development Lab Credit: 0.75

Sessional works based on SWE 4535.